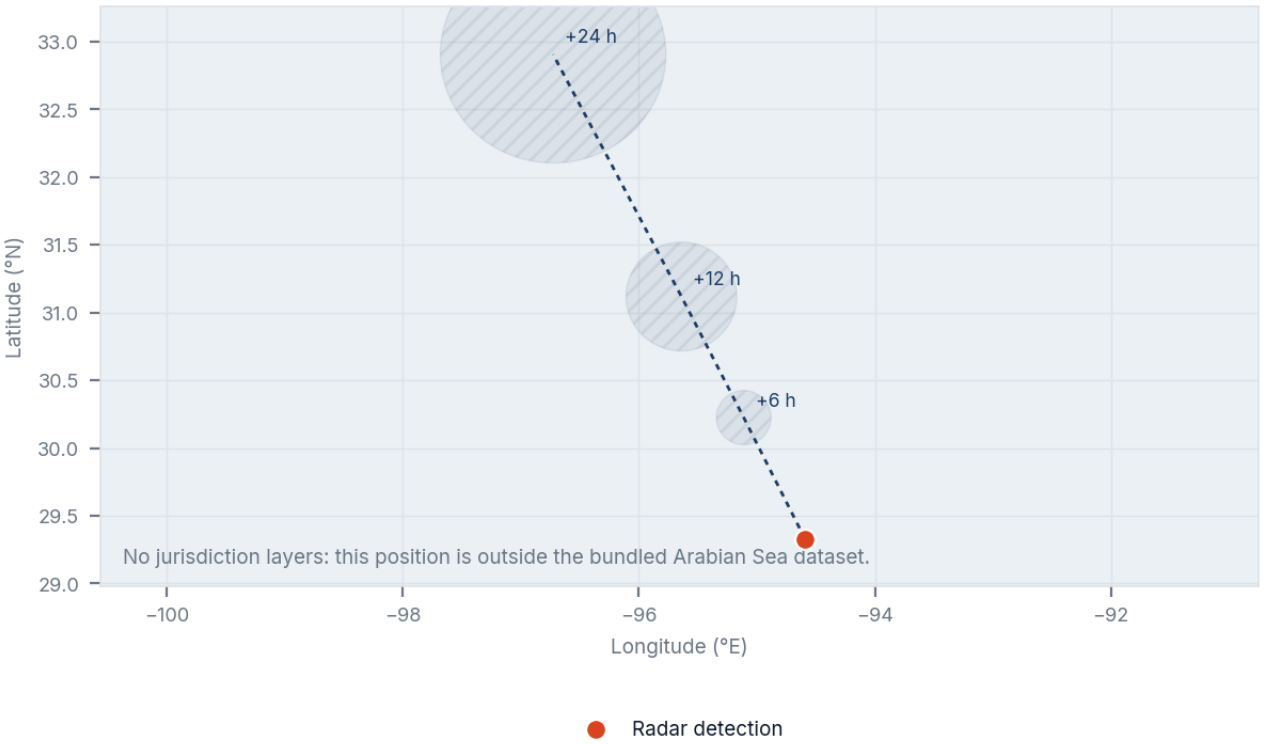
Radar contact: unmatched, AIS gap likely

Case	REAL-209848000_20240811T001908	Flagged (UTC)	2026-10-11 12:38
Threat tier	MEDIUM	Position	29.3304, -94.5926
Hypothesis	0.60 dark-vessel probability	Data	Real Sentinel-1 chip, AIS withheld as a test
Decided by	the rule-based risk scorer	Human review	not required
Target	209848000_20240811T001908-02	Mission	MSN-20261011-180807

A 180 m radar target at (29.33, -94.59) has no matching AIS broadcast. The rule-based score is 0.60 against a dark-evasion line of 0.70, giving threat tier MEDIUM.

Written by the report engine from the stored evidence. No language model was used for this text.

Position



Solid marker: radar detection. Triangle: AIS position, if any. Dashed line and hatched circles: dead-reckoning track with growing uncertainty. Dotted ellipses: ship-to-ship corridors.

Decision

THREAT TIER: MEDIUM

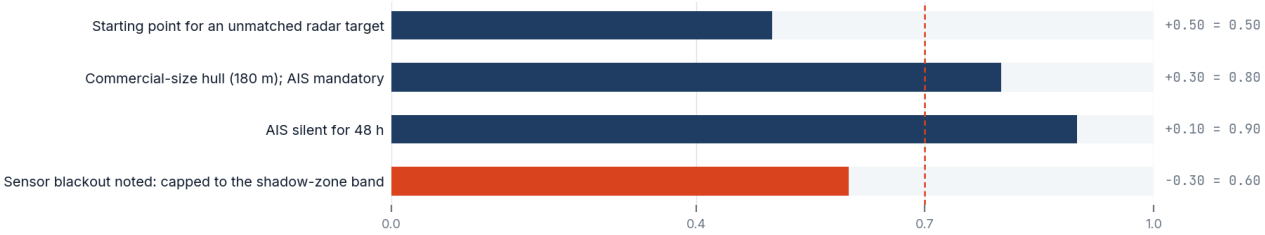
Target SAR-209848000_20240811T001908-02 was detected by Sentinel-1 SAR at (29.3304, -94.5926) with an estimated hull of 180 m × 45 m (detector confidence 0.86). No AIS broadcast could be correlated with it within the spatio-temporal gate.

The deterministic scorer classifies the target as AIS_SHADOW_ZONE with a dark-vessel probability of 0.60 and an overall anomaly score of 0.41, giving threat tier MEDIUM.

The position lies outside the bundled EEZ/STS dataset, so no jurisdictional reasoning was applied. Sanctions screening: no registry match. The run was degraded (3 condition(s)); see the agent-workflow section.

0.60	0.41	ais shadow zone
Dark-vessel probability (higher means more suspicious)	Anomaly score: 0.60 × probability, plus sanctions and EEZ terms	Classification

Each bar shows the running probability after one rule. The dashed line is 0.70, where an unmatched target is called a deliberate dark-vessel case.



WHAT IT DOES

Starts from a neutral prior for a radar target with no AIS match, then adds or subtracts fixed amounts for hull size, distance offshore, ship-to-ship corridor proximity and how long AIS has been silent.

WHAT IT TELLS YOU

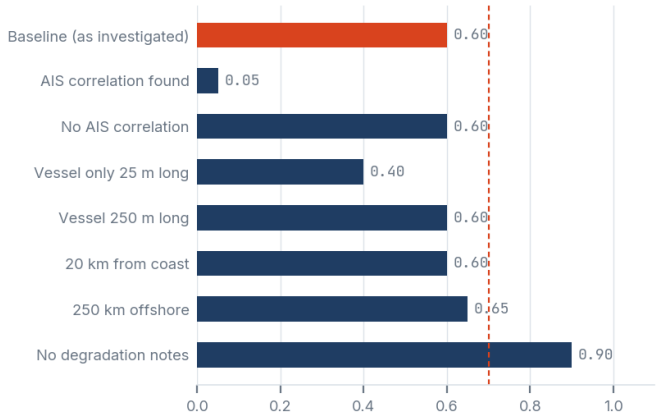
How strongly the evidence points to AIS being off on purpose. It is a rule-based judgement with documented thresholds, not a calibrated probability, so treat the number as a ranking aid.

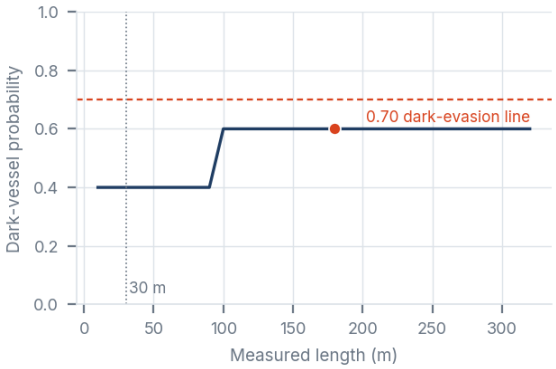
METHOD src/analytics/risk_scorer.py · reports/risk_scorer_validation.md

What if the evidence were different?

CONTEXT

Each row re-runs the production scorer with one input changed. It shows how fragile or robust the call is.



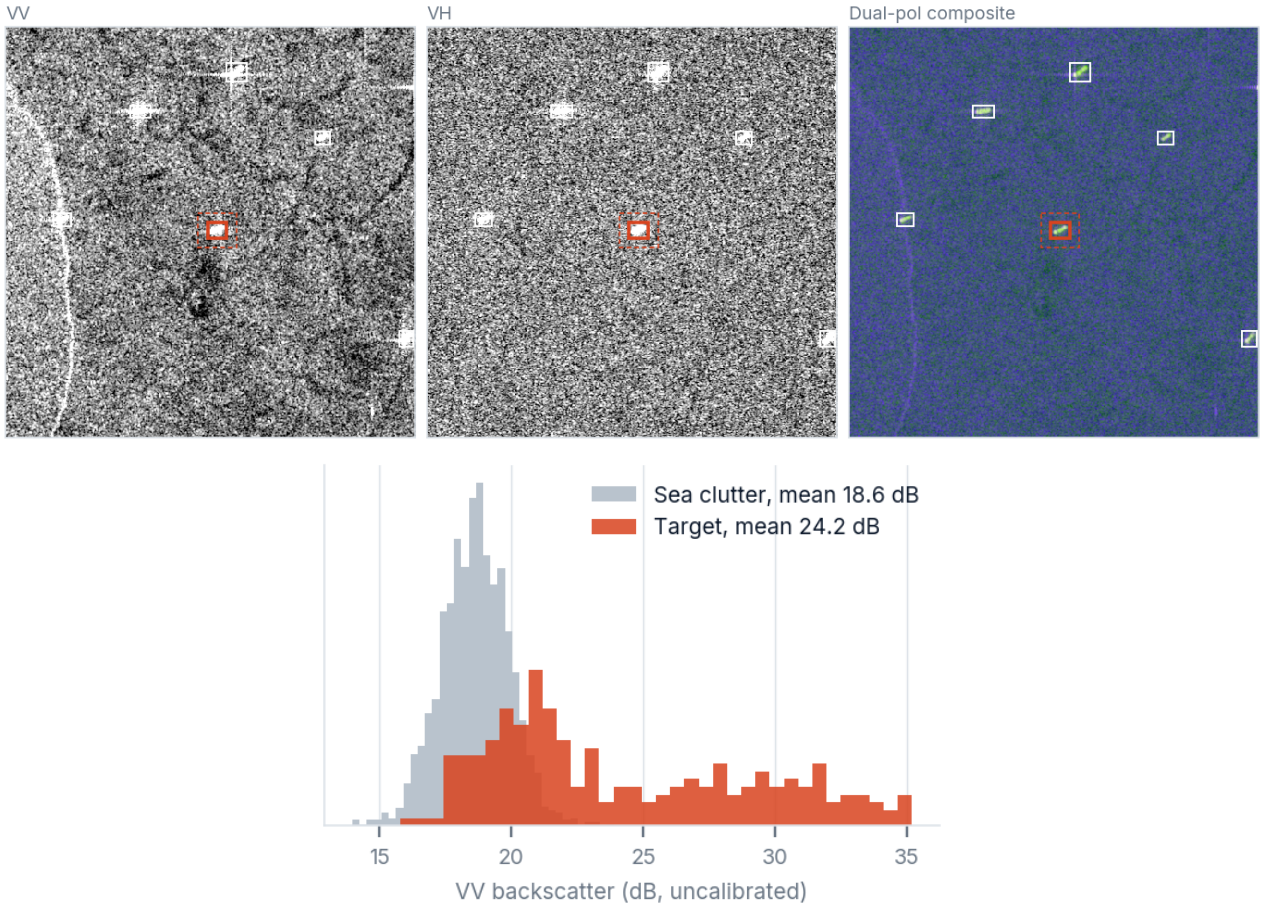


Scenario	Probability	Classification	Input changed
Baseline (as investigated)	0.60	ais shadow zone	—
AIS correlation found	0.05	correlated benign	`fusion.is_spatially_correlated = True`
No AIS correlation	0.60	ais shadow zone	`fusion.is_spatially_correlated = False`
Vessel only 25 m long	0.40	ais shadow zone	`sar.length_m = 25`
Vessel 250 m long	0.60	ais shadow zone	`sar.length_m = 250`
20 km from coast	0.60	ais shadow zone	`distance_to_coast_km = 20`
250 km offshore	0.65	ais shadow zone	`distance_to_coast_km = 250`
No degradation notes	0.90	deliberate dark evasion	`degradation_notes = []`

Radar detection

CONTEXT

The detector found a ship-sized target about 180 m long and 45 m wide, with confidence 0.86, standing 5.6 dB above the surrounding sea.



Length 180 m Beam 45 m Confidence 0.86 Contrast 5.6 dB

WHAT IT DOES

A neural detector finds ships in the dual-polarisation radar image. Length and beam come from the size of the detected box at 10 m per pixel; contrast comes from comparing the target with a 15-pixel ring of sea around it.

WHAT IT TELLS YOU

That something ship-sized was at this place and time, whatever its transponder says. Size is good to about 10 m, and the heading read from the box is coarse.

Detection	Confidence	Length	Beam	Contrast
9848000_20240811T001908-00	0.88	190 m	170 m	4.8 dB
9848000_20240811T001908-01	0.87	150 m	41 m	6.1 dB
9848000_20240811T001908-02	0.86	180 m	45 m	5.6 dB
9848000_20240811T001908-03	0.85	130 m	35 m	5.9 dB
9848000_20240811T001908-04	0.84	130 m	48 m	7.0 dB
9848000_20240811T001908-05	0.83	200 m	39 m	8.5 dB

METHOD src/models/sar_detector.py · reports/feature_math_validation.md

AIS correlation

FIRED

AIS Transmission Status: NON_TRANSMITTING (Zero active AIS signals received). Outage duration: 48.0 hrs. Suspected dark vessel.

Measure	Value
Distance, radar to AIS	n/a: no AIS candidate
Time difference	n/a: no AIS candidate
Length mismatch	—
Heading difference	—
Identity score	n/a: no AIS candidate

WHAT IT DOES

Looks for an AIS broadcast within 15 km and 60 minutes of the radar position, then compares distance, time, length and heading between the two.

WHAT IT TELLS YOU

A match explains the radar target as an ordinary transmitting ship. No match means only that no transmitter was found: the vessel may be exempt, outside receiver coverage, or hiding.

METHOD src/agents/tools.py, correlate_sar_ais

Position and jurisdiction

COULD NOT CHECK

Geospatial Context: Position (29.3304, -94.5926) lies outside the bundled Arabian Sea EEZ/STS dataset. No jurisdiction, port-proximity or corridor analysis was performed.

The bundled zone and corridor data cover the Arabian Sea only. Distances for this position would be measured to features thousands of kilometres away, so none are reported and the score ignores them.

WHAT IT DOES

Places the position against the bundled maritime zones and corridors, and measures distance to coast, EEZ boundary, port and ship-to-ship transfer corridor.

WHAT IT TELLS YOU

Which state's rules apply, and whether the position fits a known transfer pattern. Outside the dataset it says nothing at all.

METHOD src/geospatial/eez_checker.py

Sanctions screening

NORMAL

Sanctions Status: CLEAN. None

WHAT IT DOES

Compares the identifiers available (IMO, MMSI, name) against the OFAC SDN and UN consolidated lists, loaded from the official files.

WHAT IT TELLS YOU

A hit is a lead for the registry owner, not proof of identity. With no identifier, as for an unmatched radar target, the screen cannot say anything.

METHOD src/data/sanctions_parser.py · list file hashes in the appendix

Forecast

CONTEXT

Horizon	Latitude	Longitude	Uncertainty	Time (UTC)
+6 h	30.225	-95.111	±22.3 km	2024-08-11 06:19
+12 h	31.119	-95.639	±44.7 km	2024-08-11 12:19
+24 h	32.907	-96.726	±89.3 km	2024-08-12 00:19

Kinematic dead-reckoning based on SOG 10.0 kts, heading 333.4 deg with 12% expanding variance cone. No AIS speed exists for this target, so a default speed is assumed along the radar-estimated heading.

WHAT IT DOES

Projects the position forward at constant speed and heading, widening a circle of uncertainty for drift and speed error.

WHAT IT TELLS YOU

Where to look next. It is not where the ship will be: it assumes straight, steady travel and says nothing about intent.

METHOD src/analytics/risk_scorer.py, extrapolate_trajectory · reports/trajectory_validation.md

Agent workflow

DEGRADED



Step	Node	Status	Time	What happened
1	MissionController	DEGRADED	1.0 ms	Mission initialized; geospatial context unavailable (outside dataset coverage).
2	SARAISCorrelator	DEGRADED	0.1 ms	AIS telemetry feed unavailable or zero broadcasts in spatio-temporal window: degraded to dark vessel radar-only mode.
3	FeatureEngineerNode	SUCCESS	0.2 ms	Successfully extracted 5-domain mathematical feature vector.
4	SanctionsScreenerNode	SUCCESS	1039.1 ms	SCREEN_OFAC_UN_SANCTIONS

Step	Node	Status	Time	What happened
5	ForensicAssessorNode	SUCCESS	0.1 ms	Assessed risk: AIS_SHADOW_ZONE (Anomaly Score: 0.41). HITL=False.
6	ReportPreparerNode	SUCCESS	0.1 ms	Compiled bulletin MAR-INTEL-20261011-908-02 with threat tier MEDIUM.

Where evidence was missing

— AIS transponder feed offline (satellite blind zone)

— Position is outside the bundled EEZ/STS dataset coverage: geospatial risk factors were not applied.

— AIS telemetry feed unavailable or zero broadcasts in spatio-temporal window: degraded to dark vessel radar-only mode.

WHAT IT DOES

Seven software steps run in order: mission set-up, radar-to-AIS matching, feature engineering, sanctions screening, risk assessment, an optional human step, and report writing.

WHAT IT TELLS YOU

Which evidence each step used, how long it took, and where data was missing or the run fell back to a simpler mode.

METHOD src/agents/graph.py · reports/agentic_ai_audit.md

Human review and audit trail

NOT REQUIRED

The risk gate did not require a person to decide this case.

The audit ledger holds 2 entries and its hash chain verifies. Head: 9a51c7586406ef23d80fc6a5.

Entry	Time (UTC)	Action	By
1	2026-10-11 12:38:08	INVESTIGATION_COMPLETED	SYSTEM

Legal basis

CONTEXT

Statutory text comes from a fixed catalogue; a language model never writes it.

SOLAS Chapter V Regulation 19

Mandates continuous AIS broadcasts for all commercial vessels >=300 GT on international voyages and all cargo vessels >=500 GT on domestic routes.

Why it applies: Radar length 180 m with no AIS correlation: AIS carriage is mandatory for commercial vessels of this size.

https://www.imo.org/en/OurWork/Safety/Pages/AIS.aspx

Considered and not cited

Provision	Why it was not cited
UNCLOS Article 73	position is not inside a mapped sovereign EEZ or the vessel is transmitting.
UNCLOS Article 92	no nationality/flag irregularity evidence in the inputs.
UNCLOS Article 110	right-of-visit grounds (stateless / piracy) are not evidenced.
OFAC SDN / IEEPA Statutory Sanctions	no confirmed sanctions match.

Explanations, in order of how well the evidence fits

#	Explanation	Fit	Evidence for	Evidence against
1	Counterfactual test artefact (AIS was withheld on purpose)	certain	This case uses a real radar chip whose true AIS (MMSI 209848000, NISI) was hidden.	nothing recorded
2	AIS switched off on purpose	medium	Commercial hull (180.0m) legally mandated to broadcast AIS under SOLAS V/19.; Geospatial context unavailable: position is outside the regional EEZ/STS dataset, so offshore and STS factors were not applied.; Active AIS outage duration exceeds 48.0 hours.; Sensor network degradation/blackout detected; evidence consistent with AIS shadow zone.	A radar detection alone cannot show intent.
3	AIS coverage gap or satellite blind zone	medium	AIS transponder feed offline (satellite blind zone); AIS telemetry feed unavailable or zero broadcasts in spatio-temporal window: degraded to dark vessel radar-only mode.	No independent AIS-coverage map is bundled, so coverage cannot be verified here.
4	Legitimately small craft exempt from AIS	low	Measured length 180 m is above the exemption threshold.	Hull is far larger than craft normally exempt under SOLAS Ch. V Reg. 19.; Length is estimated from an axis-aligned detector box at 10 m pixel spacing (±10 m quantisation).
5	AIS spoofing or position error (vessel is transmitting, but wrongly)	low	Nearest AIS report is 999.0 km away.	Spoofing cannot be distinguished from absence using one SAR snapshot.
6	Detector false positive (sea clutter, platform, wake)	low	Detector confidence 0.86.; Target-to-clutter ratio 5.6 dB.	Confidence and contrast are well above typical clutter returns.

WHAT WOULD SETTLE IT

— None required — this is the intended test condition, not an observed evasion.
— A second, independent AIS receiver feed or a satellite pass at the same time.
— Query terrestrial/satellite AIS receiver coverage for the acquisition time.
— Optical or FLIR confirmation, or a higher-resolution SAR acquisition.
— Multi-pass SAR + AIS track continuity analysis.

Checks that came back normal or could not run

Check	Result	Here
AIS correlation	Fired	No AIS broadcast matched the radar target within the 15 km / 60 min gate.
Sanctions registries	Normal	No OFAC or UN entry matched the identifiers available.
Detector confidence	Normal	Confidence 0.86; target-to-clutter ratio 5.6 dB.
Jurisdiction data	Could not check	Position is outside the bundled Arabian Sea EEZ and corridor dataset.
Evidence completeness	Degraded	3 condition(s) degraded the run; see the workflow section.
Human review	Not required	The risk gate did not require a person to decide.
Audit ledger	Normal	2 entries; hash chain verified.

What this assessment cannot tell you

— A radar detection alone cannot show that a vessel is deliberately hiding; the output is a probability, not a finding of fact.

— Length, beam and heading are estimated from detector boxes at 10 m pixel spacing and carry material uncertainty.

— The risk score is a rule-based heuristic, not a calibrated statistical model; thresholds are documented in §9.

— Historical features (flag/name changes, route anomaly, past STS) are heuristic placeholders unless an AIS history was supplied.

— Single-snapshot assessment: no multi-pass confirmation, wake analysis or Doppler speed estimate.

— The position is outside the bundled Arabian Sea EEZ/STS dataset; no jurisdictional or corridor reasoning was applied.

— AIS evidence is absent or withheld.

Evidence ledger

Every engineered feature behind this assessment. Values marked n/a had no AIS candidate to compute from.

Feature	Class	Value	Meaning
length_m	Radar	180.00	Estimated target length in meters
width_m	Radar	44.70	Estimated target beam in meters
aspect_ratio	Radar	4.03	Ratio of length to beam (L/B) (length_m / width_m)
detection_area_m2	Radar	6034.50	Target pixel area converted to square meters
orientation_deg	Radar	333.40	Major axis orientation in degrees
peak_backscatter_db	Radar	35.17	Maximum sigma-nought intensity inside target mask
mean_backscatter_db	Radar	24.23	Mean sigma-nought intensity inside target mask
background_clutter_db	Radar	18.62	Estimated background clutter sigma-nought
target_to_clutter_ratio_db	Radar	5.61	Target contrast over background clutter (mean target dB - mean clutter dB)
backscatter_variance	Radar	—	Variance of pixel values across the ship target
compactness	Radar	1.96	Perimeter^2 / (4 * pi * Area) (perimeter^2 / (4 * pi * area))
eccentricity	Radar	0.97	Second moment ellipse eccentricity
incidence_angle_deg	Radar	—	Radar incidence angle
polarization_ratio	Radar	—	VH / VV backscatter ratio if dual-pol available
mean_speed_knots	AIS track	n/a	Mean Speed Over Ground across track window
speed_variance	AIS track	n/a	Variance of SOG across track window
max_speed_knots	AIS track	n/a	Maximum recorded SOG
speed_acceleration_kph2	AIS track	n/a	Rate of speed change (acceleration/deceleration)
heading_variance	AIS track	n/a	Circular variance of vessel heading/course (1 - |sum of unit heading vectors| / N (circular variance))
mean_turn_rate_deg_min	AIS track	n/a	Rate of course change per minute
trajectory_curvature	AIS track	n/a	Ratio of actual path length to endpoint displacement (path length / endpoint displacement)
total_distance_km	AIS track	n/a	Total distance travelled along observed AIS track
ais_gap_count_30d	AIS track	n/a	Total number of AIS transmission outages in past 30 days
longest_ais_gap_hours	AIS track	n/a	Maximum continuous AIS outage duration
mean_ais_gap_hours	AIS track	n/a	Average duration of AIS outages

Feature	Class	Value	Meaning
current_gap_duration_hours	AIS track	n/a	Elapsed time since last valid AIS broadcast
loitering_duration_hours	AIS track	n/a	Duration spent drifting at speed < 3 knots
distance_to_shore_km	Geography	13463.90	Straight-line distance to closest land boundary
distance_to_nearest_port_km	Geography	13286.17	Distance to closest recognized commercial port
distance_to_eez_boundary_km	Geography	13093.90	Distance to nearest national Exclusive Economic Zone boundary
is_inside_eez	Geography	no	Binary indicator: located inside sovereign EEZ
is_inside_mpa_or_rfmo	Geography	no	Binary indicator: located in protected marine area
proximity_to_sts_corridor_km	Geography	13216.00	Distance to designated ship-to-ship transfer zone
proximity_to_nearest_vessel_km	Geography	—	Distance to nearest concurrent vessel in area
vessel_density_local_km2	Geography	—	Local vessel traffic density in 25km radius
prior_sanctions_involvement	History	no	Flagged in historical maritime illicit trade reports
flag_change_count_12m	History	0	Number of flag changes in past 12 months (Flag hopping indicator)
name_change_count_24m	History	0	Number of vessel name changes in past 24 months
historical_loitering_events_6m	History	0	Count of past suspicious slow-speed drift events
historical_sts_encounters_6m	History	0	Count of past rendezvous within 500m of another tanker
route_anomaly_score	History	0.00	Deviation score against historical traffic lanes
spatial_discrepancy_km	Fusion	n/a	Distance between SAR target position and extrapolated AIS position (haversine(SAR position, AIS position))
temporal_difference_minutes	Fusion	n/a	Time delta between SAR acquisition and closest AIS broadcast (|t_SAR - t_AIS| in minutes)
dimension_mismatch_ratio	Fusion	—	|SAR_length - AIS_length| / AIS_length (|L_SAR – L_AIS| / L_AIS)
heading_discrepancy_deg	Fusion	—	Angular difference between SAR orientation and AIS COG (min(|SAR heading - COG|, 360 - |SAR heading - COG|))
speed_consistency_score	Fusion	0.00	Consistency of SAR Doppler/wake speed vs AIS SOG
predicted_ais_position_error_km	Fusion	—	Extrapolation uncertainty radius from AIS dead-reckoning
is_spatially_correlated	Fusion	no	Binary: whether SAR target is explained by a known AIS vessel

Feature	Class	Value	Meaning
candidate_identity_score	Fusion	n/a	Composite correlation score across all dimensions (weighted spatial + temporal + dimension agreement)

Where the data comes from

Input	Provenance
Radar imagery	REAL
Radar measurements	DERIVED
AIS	UNAVAILABLE
Sanctions registries	REAL
Jurisdiction (EEZ / STS)	UNAVAILABLE
Historical features	HEURISTIC
Risk classification	HEURISTIC
Trajectory forecast	DERIVED

Asset	Provenance	Count	Source
Sentinel-1 SAR chips (VV+VH)	REAL	200	Hugging Face mshaya/sar-ship-gulf (Sentinel-1 IW GRDH dual-pol chips with co-registered AIS) Acquired 2024-08-11 to 2024-08-13; located off the US Gulf coast (outside the Arabian Sea EEZ dataset).
Co-registered AIS for each chip	REAL	200	Hugging Face mshaya/sar-ship-gulf (Sentinel-1 IW GRDH dual-pol chips with co-registered AIS) MMSI, name, type, true length, SOG and heading at acquisition time.
OFAC SDN list	REAL	—	U.S. Treasury OFAC Official designation list parsed into an in-memory IMO/MMSI/name index.
UN Security Council consolidated list	REAL	—	UN Security Council Official designation list parsed into an in-memory IMO/MMSI/name index.
Sanctioned vessels indexed	REAL	1563	OFAC + UN
EEZ / maritime zone polygons	REAL	3	Regional UNCLOS zones (Arabian Sea only). Exact upstream dataset is not recorded in the repo. Chips outside this region get no EEZ/STS context.
Large SAR detection table	SYNTHETIC	35000	Generated by src/data/download_high_volume_bundle.py Filename says 'real' but rows are procedurally generated.
AIS traffic table	SYNTHETIC	103120	Generated by src/data/download_high_volume_bundle.py Filename says 'real' but rows are procedurally generated.

Asset	Provenance	Count	Source
Benchmark scenarios	SYNTHETIC	5	Hand-written test fixtures Used to exercise each decision path deterministically.
YOLO SAR vessel detector	REAL	—	Trained weights
Qwen 2.5 maritime model (merged LoRA)	REAL	—	Fine-tuned on rule-generated report examples Loadable: Used only to draft report prose; numbers and citations come from code.

```
OFAC SDN list: d533a38ee0a8dc1dca61574cda864d1740ec6785f1ae93f00e678fdf01654227
UN Security Council consolidated list: af812e39a32fb4d146a4dd91a2467e9647565c17b04fa3be350fe807da1dde43
```

Integrity and reproduction

Item	Value
Evidence hash (SHA-256)	`2ebd2ed75d93ae9ef2f4fa0419d5ae27f260bcdf4d740b2a5514a42850531a50`
Audit chain head	`9a51c7586406ef23d80fc6a5c1d7289cdacfbc9b7e6b02d78d72b611dd9d9537`
Software	`{"git_sha": "e9f13eb", "python": "3.12.3", "platform": "Windows-11-10.0.26200-SP0", "reportlab": "5.0.1", "matplotlib": "3.10.0", "numpy": "2.2.3"}`
Narrative	`Rule-generated narrative`

REQUEST THAT PRODUCED THIS REPORT


```
{
 "mission_id": null,
 "sar_detection": {
  "detection_id": "SAR-209848000_20240811T001908-02",
  "scene_id": "S1A_IW_GRDH_1SDV_20240811T001856_20240811T001921_055158_06B8DB_E8C5.SAFE",
  "timestamp": "2024-08-11T00:19:08.500000Z",
  "latitude": 29.33038,
  "longitude": -94.59264,
  "length_m": 180.0,
  "width_m": 44.7,
  "aspect_ratio": 4.03,
  "heading_deg": 333.4,
  "area_m2": 6034.5,
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  "mean_backscatter_db": 24.23,
  "background_mean_db": 18.62,
  "target_clutter_ratio_db": 5.61,
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   247,
   273,
   260
  ]
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 "simulate_sanctions_offline": false,
 "human_feedback": null,
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```
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 "scenario_id": "REAL-209848000_20240811T001908",
 "ground_truth": null,
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  "ais": "UNAVAILABLE",
  "classification": "HEURISTIC",
  "imagery": "REAL",
  "source": "Hugging Face mshaya/sar-ship-gulf (Sentinel-1 IW GRDH dual-pol chips with co-registered AIS)",
  "note": "Imagery and AIS are real; measurements are derived by the detector. AIS was deliberately withheld t
o simulate a dark vessel: this is a counterfactual test, not an observed evasion."
 }
}
```


Send this request to POST /api/investigate. The deterministic stages (features, scoring, forecast, sanctions) return identical values for identical inputs.